

Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

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Abstract

001 Evaluating AI/ML systems responsibly requires prac-
002 tices for testing, stress-testing, auditing, comparing,
003 and interpreting specific capabilities—not only aggre-
004 gate benchmark scores. This paper proposes such a
005 practice for spatial reasoning in vision–language mod-
006 els (VLMs): a diagnostic audit of object-relative orien-
007 tation, a capability that aggregate accuracy can hide.
008 On Visual Spatial Reasoning (VSR), moving from a
009 compact 2.2B model to a larger 7B VLM substantially
010 improves overall, depth, horizontal, and containment
011 accuracy, but orientation changes only modestly; a
012 large 2026 sparse-MoE model (MiMo-V2.5, 310B total /
013 15B active) reproduces the pattern (79.5% overall,
014 65.7% orientation). Except for structured prompt-
015 ing, which degrades performance, trained conditions
016 on both models and the large sparse-MoE zero-shot
017 model remain in a narrow 62–66% orientation band.
018 We test structured prompting, general and targeted
019 LoRA, hard negatives, visual/projector adaptation,
020 frozen and object-grounded representation probes, and
021 object-centric decomposition; none of the tested config-
022 urations reliably closes the gap, and a label-sensitivity
023 analysis (an automated audit followed by two blind
024 human re-audits; the humans agree well on the fail-
025 ure taxonomy, while clean/ambiguous exclusion masks
026 remain annotator-dependent and automated auditors
027 vary substantially in calibration) shows that the pat-
028 tern persists after excluding examples flagged as am-
029 biguous.

030 Accuracy also fails to capture relational coherence.
031 LM-side adaptation significantly improves consistency
032 over zero-shot, while hard-negative training raises
033 facing/facing-away consistency from 66.0% to 77.7%
034 with facing accuracy unchanged at 68.9%; the addi-
035 tional pooled consistency gain over LM-only is not sig-
036 nificant. Cross-dataset evaluation on SITE shows that
037 VSR-trained adaptation does not transfer cleanly: it is

neutral overall and significantly harms official spatial-
relationship reasoning (75.1%→71.6%, $p = 0.004$). A
pre-specified orientation-vocabulary slice, frozen before
SITE evaluation, is analyzed as exploratory evidence
rather than as direct replication of the VSR construct.
The results point to a persistent orientation bottleneck
in which orientation is not robustly decodable by the
tested frozen-feature readouts and relational decisions
are not always propagated coherently, while also show-
ing that task-specific spatial adaptation can transfer
poorly. Each component of the audit—the diagnos-
tic ladder, the label-sensitivity check, the accuracy–
coherence separation, and the cross-dataset transfer
protocol—is specified operationally so that it can be ap-
plied by others to other capabilities, benchmarks, and
model families.

1. Introduction

This paper contributes an evaluation practice as much
as a set of model results. We develop a protocol for
testing, stress-testing, auditing, comparing, and inter-
preting one capability across vision–language models
(VLMs), and instantiate it for object-relative orienta-
tion: facing, facing away, parallel, and perpendicular.
This capability is relevant to robotics, embodied agents,
navigation, and scene understanding, but it is easy to
hide behind an aggregate score: success at identifying
the relevant objects, inferring the relation, or answering
complementary formulations does not guarantee suc-
cess at the others.

Prior benchmarks establish that grounded spatial
reasoning remains difficult, but they leave open how
to diagnose a relation-specific failure. VSR intro-
duced a controlled natural-image probe and reported
particularly poor object-orientation performance [7];
What’sUp and GSR-Bench similarly expose failures on
basic grounded relations [6, 9]; and broader evaluations
continue to identify orientation as a difficult factor [15].

075 We therefore ask three narrower questions: does the
076 VSR orientation weakness persist in modern genera-
077 tive VLMs and at much larger scale; can prompting,
078 adaptation, or visual readouts remove it; and do accu-
079 racy, logical coherence, and cross-dataset transfer tell
080 the same story?

081 The novelty is diagnostic rather than a claim that
082 we discovered spatial difficulty. Our contribution is not
083 any individual diagnostic, but a falsification protocol
084 that holds the relation family and examples fixed while
085 testing whether an apparent capability failure survives
086 scaling and adaptation, label re-audit, representation
087 readout, logical reformulation, and cross-dataset trans-
088 fer. Prior work generally reports benchmark accuracy,
089 studies a single intervention, or proposes a consistency
090 objective; combining these views into one controlled
091 audit of the same relation family is what lets us dis-
092 tinguish a persistent relation-specific bottleneck from
093 aggregate underperformance, annotation artifacts, or
094 benchmark-specific specialization.

095 In our VSR experiments, the distinction between
096 relation families is striking. Moving from SmolVLM2-
097 2.2B-Instruct to the larger Qwen2-VL-7B-Instruct in-
098 creases zero-shot overall accuracy from 74.0% to 80.9%.
099 Horizontal reasoning rises from 70.2% to 84.8%, depth
100 from 68.9% to 75.2%, and containment from 83.4%
101 to 89.3%. Orientation, by contrast, moves only from
102 62.8% to 63.5%. A large 2026 sparse-MoE model
103 (MiMo-V2.5, 310B total / 15B active parameters) re-
104 produces the same relation-specific pattern: 79.5%
105 overall accuracy yet 65.7% on orientation, again in-
106 side the 62–66% band. We then evaluate interven-
107 tions spanning prompting, language-side adaptation,
108 targeted data construction, visual-side adaptation, rep-
109 resentation probing, and explicit object-centric decom-
110 position: structured prompting significantly hurts; gen-
111 eral and targeted LoRA improve aggregate accuracy
112 but do not remove the persistent orientation weakness;
113 hard negatives improve targeted relational coherence
114 without a significant orientation-accuracy gain; and the
115 tested visual-side LoRA configurations do not outper-
116 form the language-side control.

117 As a sensitivity analysis, removing examples flagged
118 by the automated audit raises orientation accuracy
119 (65.7%→78.5% for 7B General LoRA, 66.4%→76.6%
120 for hard-negative LoRA on the strictest 107-example
121 subset) but does not eliminate error (2B structured
122 shows no gain). The exclusion sets are annotator-
123 dependent: two blind human re-audits under the
124 same taxonomy agree strongly with each other on the
125 failure taxonomy (Krippendorff’s $\alpha=0.81$) while the
126 clean/ambiguous flags agree only moderately with the
127 automated audit ($\kappa=0.36$ on the 48 shared persistent-

failure cases); details are in the supplementary. Be-
cause the removed examples were drawn from persis-
tent failures, part of this gain is expected by construc-
tion.

A second failure mode emerges from logical consis-
tency. The zero-shot 7B model is consistent on only
36.9% of complementary facing/facing-away pairs. LM-
side LoRA substantially improves coherence, and hard-
negative training raises facing consistency to 77.7%
while facing accuracy remains unchanged at 68.9%.
The extra hard-negative gain is not significant in the
pooled strict-family comparison, yet accuracy stays un-
changed while consistency moves: the two metrics cap-
ture different behavior.

Finally, we test cross-dataset transfer. The 7B
model is strong on SITE’s official spatial-relationship-
reasoning category (75.1% raw; 59.2% chance-adjusted
accuracy), yet the VSR-trained adapter does not
generalize cleanly: it is statistically neutral overall
(54.2%→53.8%, $p = 0.66$) and significantly degrades
the official category (75.1%→71.6%, $p = 0.004$). A
pre-specified, non-official orientation-vocabulary slice
is much weaker in the aggregate (48.3% raw; 24.0%
chance-adjusted) and shows no positive transfer, but—
as we detail below—its composition limits what it can
establish, and we therefore do not treat SITE as inde-
pendent confirmation of the VSR orientation construct.

Our contributions are four reusable components of
an evaluation protocol:

- A fine-grained diagnostic ladder. Across ten VSR
conditions, two differently sized models, and a large
zero-shot sparse-MoE model, several spatial fami-
lies improve substantially while object-relative ori-
entation remains unusually resistant. The ladder—
scaling, prompting, language-side adaptation, tar-
geted data construction, visual-side adaptation, rep-
resentation probing, and explicit decomposition—is
defined operationally so it transfers to other bench-
marks and relation families.
- A robustness and sensitivity audit. A label-
sensitivity analysis (automated labels with two inde-
pendent human re-audits), frozen/object-grounded
probes, visual-side adaptation, and object-centric
decomposition constrain purely annotation-based or
single-component explanations. The audit practice
(flagged-example exclusion sets, decodability checks
against majority baselines) applies to any bench-
mark whose labels or examples may be ambiguous.
- An accuracy-coherence separation protocol. Com-
plementary-relation tests on VSR’s strict
relation pairs show that, for the tested conditions,
relational consistency can change substantially
while task accuracy remains fixed. The protocol

181	measures consistency against verdict-pattern base	232
182	rates, so it can be applied wherever complementary	233
183	formulations of the same relation exist.	234
184	• A cross-dataset transfer protocol. SITE shows	235
185	strong official spatial-relationship performance,	236
186	while VSR-trained LoRA transfers poorly and sig-	237
187	nificantly harms the official SITE category; the	238
188	orientation-vocabulary slice is analyzed as ex-	239
189	ploratory evidence. The protocol (pre-specified	240
190	subsets, chance-adjusted accuracy, paired transfer	241
191	tests) is model- and benchmark-agnostic and is re-	242
192	ported with negative as well as positive transfer.	243
193	The protocol is deliberately modular: each compo-	
194	nent can be applied independently to other capabilities,	
195	benchmarks, and model families. The supplementary	
196	specifies the prompts, subsets, decision rules, and re-	
197	leased artifacts needed to reproduce the audit.	
198	2. Related Work	
199	Benchmarks for grounded spatial reasoning. VSR in-	
200	troduced a controlled natural-image benchmark with	
201	66 spatial relations and identified facing and parallel as	
202	difficult relations for the VLMs studied at the time [7].	
203	What'sUp controls object identity while varying spatial	
204	configuration and reports a substantial human-model	
205	gap [6]; GSR-Bench broadens grounded evaluation	
206	across models and scales [9]. SITE extends the scope	
207	to single-image, multi-image, and video settings and	
208	organizes tasks by dimensions such as orientation, in-	
209	trinsic/extrinsic relations, and static/dynamic reason-	
210	ing [15]. SpatialScore and Mind the Gap further show	
211	that spatial competence remains fragmented [10, 16].	
212	We use VSR not to re-establish that spatial reasoning	
213	is hard, but to turn its orientation signal into a con-	
214	trolled diagnostic target and use SITE to test transfer.	
215	Diagnosing and improving spatial failures. Several	
216	lines of work move beyond aggregate accuracy.	
217	AdaptVis relates successful VSR reasoning to attention	
218	on relevant object locations [2], while CREG shows that	
219	attribution evidence in Qwen2-VL can reflect image lay-	
220	out rather than the queried relation [11]. When More Is	
221	Less finds that spatial cues, commonsense information,	
222	and reasoning instructions can help or hurt depend-	
223	ing on the model and setup [1]. Other work probes	
224	visual-concept decodability [13], separates perception	
225	from reasoning in spatial planning [17], or supplies ge-	
226	ometric inputs for 3D spatial understanding [3]. These	
227	studies motivate our combination of frozen/readout di-	
228	agnostics with adaptation and object-centric decompo-	
229	sition, but do not provide the same end-to-end audit	
230	of one relation family across accuracy, coherence, label	
231	sensitivity, and transfer.	
	Taken together, prior work establishes the problem	232
	and offers several useful diagnostic lenses, but leaves	233
	a methodological gap: benchmark accuracy, represen-	234
	tation access, logical consistency, annotation sensitiv-	235
	ity, and transfer are usually reported separately. We	236
	close that gap with a single controlled diagnostic lad-	237
	der, with analysis choices and external-evaluation sub-	238
	sets frozen where specified, applied to the same rela-	239
	tion family. Our claim is consequently not to discover	240
	spatial difficulty, but to make a persistent orientation	241
	weakness measurable, falsifiable, and reusable as an	242
	evaluation protocol.	243
	Adaptation, specialization, and transfer. Parameter-	244
	efficient adaptation such as LoRA [5] can improve a	245
	benchmark without improving the underlying capabil-	246
	ity. We therefore compare general, targeted, hard-	247
	negative, projector, and vision-plus-projector updates,	248
	then evaluate the strongest language-side adapter on	249
	an untouched benchmark. This design distinguishes	250
	within-benchmark specialization from transfer: VSR	251
	adaptation is neutral overall on SITE, does not im-	252
	prove the orientation subset, and significantly degrades	253
	SITE's official spatial-relationship category. Our em-	254
	phasis on negative transfer complements prior bench-	255
	mark work by treating transfer failure as an evaluation	256
	result rather than an omitted outcome.	257
	Representation probing and logical consistency.	258
	Probes can test whether a signal is accessible from	259
	frozen representations, but probe capacity can itself	260
	learn the task, requiring control baselines and cautious	261
	interpretation [4]. Consistency-oriented objectives	262
	such as SAGE use geometric and linguistic dualities	263
	to train spatial reasoning [8]. We instead use probing	264
	and complementary-pair consistency as independent	265
	diagnostics: the former asks what the tested readouts	266
	can decode, while the latter asks whether answers	267
	to logically linked statements agree, so accuracy and	268
	coherence can diverge as diagnostic axes.	269
	3. Experimental Setup	270
	3.1. Benchmarks	271
	VSR. We use VSR [7] as the primary diagnostic	272
	benchmark. The random-split test set contains 2,195	273
	examples. The benchmark defines 64 relation labels;	274
	61 appear in the evaluated test split. The orientation	275
	family contains facing, facing away from, parallel to,	276
	and perpendicular to, totaling 137 test examples. All	277
	main VSR results use the same held-out test set and	278
	deterministic True/False parsing.	279

280 SITE. We use SITE [15] for cross-dataset transfer
281 evaluation only and do not train on SITE. The image
282 evaluation contains 2,591 examples: 1,368 single-image
283 and 1,223 multi-image examples. Our primary SITE
284 subset is the official spatial-relationship-reasoning cate-
285 gory ($n = 993$). We additionally report a transparently
286 defined, non-official keyword-derived orientation subset
287 ($n = 2,024$), analyzed as exploratory evidence rather
288 than as direct replication of the VSR construct. SITE
289 results use the benchmark’s native multiple-choice for-
290 mat and chance-adjusted accuracy (CAA) in addition
291 to raw accuracy.

292 3.2. Models and Conditions

293 Our compact model is SmolVLM2-2.2B-Instruct, and
294 our larger controlled-study model is Qwen2-VL-7B-
295 Instruct [14]. We retain Qwen2-VL-7B as the con-
296 trolled anchor because it supports the complete local
297 ladder—the same base model across prompting, LoRA,
298 visual-side adaptation, probing, and decomposition—
299 rather than mixing intervention effects with an uncon-
300 trolled architecture change. We evaluate ten canon-
301 ical VSR conditions: 2B zero-shot, 2B structured
302 prompting, 2B General LoRA, 2B Targeted LoRA,
303 7B zero-shot, 7B General LoRA, 7B Targeted LoRA,
304 7B Hard-Neg LoRA, 7B Projector LoRA, and 7B
305 Vision+Projector LoRA. LoRA [5] is used for all
306 parameter-efficient adaptation conditions. We addi-
307 tionally evaluate a large zero-shot sparse-MoE model,
308 MiMo-V2.5 [12] (310B total / 15B activated param-
309 eters, dedicated 729M-parameter vision encoder, 2026),
310 served through its public API with the identical
311 frozen prompt, thinking disabled, and greedy decoding
312 (App. A). Its role is external validation of the diagnosed
313 pattern; it is not fine-tuned.

314 General LoRA uses a proportional sample across re-
315 lation families. Targeted LoRA allocates most training
316 examples to weak orientation/depth/horizontal fami-
317 lies with rehearsal from the remaining relations. Hard-
318 negative LoRA introduces audited complementary ori-
319 entation examples, primarily facing/facing-away inver-
320 sions. Vision-side conditions adapt either the multi-
321 modal projector or upper vision blocks together with
322 the projector.

323 3.3. Metrics and Statistical Tests

324 For VSR we report accuracy overall and by spatial fami-
325 ly. For SITE we report raw accuracy and CAA. Paired
326 condition comparisons use McNemar tests when ap-
327 plicable. Clean-label orientation analysis reports the
328 full orientation set and progressively stricter audited
329 subsets. Logical consistency is evaluated by construct-
330 ing complementary statements for strict relation pairs

(left/right, front/behind, facing/facing-away) and test-
ing whether the model assigns opposite verdicts as re-
quired. Parallel/perpendicular is treated separately as
a soft complement because both relations can be false
for oblique objects.

512 4. Main Results

513 4.1. Orientation Remains Weak Across Two Differ- 514 ently Sized VLMs

515 Table 1 shows the central pattern. Moving from the 2B
516 model to the 7B model lifts zero-shot overall accuracy
517 by 6.9 pp, horizontal by 14.6 pp, depth by 6.3 pp, and
518 containment by 5.9 pp, but orientation by only 0.7 pp.
519 General 7B LoRA further raises overall accuracy to
520 84.7%, yet orientation reaches only 65.7%. Across
521 trained conditions other than structured prompting,
522 orientation remains in a narrow 62–66% band. The
523 large sparse-MoE zero-shot model (MiMo-V2.5, evalu-
524 ated identically, App. A) falls in the same band: 79.5%
525 overall and 65.7% orientation, with the orientation fam-
526 ily again its weakest relation family (per-family table
527 in the supplementary). The relation-specific pattern is
528 therefore not an artifact of the 2B/7B scale range. The
529 MiMo-V2.5 row is also robust to output-format parsing:
530 all 137 orientation outputs were parseable (orientation
531 unchanged at 65.7% parseable-only), and parseable-
532 only overall accuracy is 82.1% vs. 79.5% with the 69
533 non-conforming outputs counted wrong (App. G).

534 To quantify the family-relative claim directly,
535 a paired bootstrap over the 2,195 shared exam-
536 ples (10,000 resamples) confirms the horizontal-vs-
537 orientation scaling contrast is robust (+13.9 pp; 95%
538 CI [+2.9, +24.8]; positive in 99.3% of resamples), with
539 depth (+5.5 pp) and containment (+5.1 pp) direction-
540 ally positive but not individually significant (App. D).

541 Structured prompting is a negative result: overall
542 accuracy falls from 74.0% to 68.3% at 2B, while ori-
543 entation falls from 62.8% to 53.3%. A paired exact
544 McNemar test on the full test set confirms that the
545 structured prompt is worse than the minimal baseline
546 ($p \approx 2.2 \times 10^{-7}$). Consistent with recent analyses of
547 information injection on VSR, where additional spatial
548 cues or reasoning instructions can help or hurt depend-
549 ing on the model and setup [1], this result is specific
550 to the tested 2B condition and prompt; it does not
551 establish that prompting generally fails.

552 4.2. Clean-Label Sensitivity Analysis

553 An automated audit (LLM-generated labels; the frozen
554 v1 table) identified ambiguous or questionable orienta-
555 tion examples, so we repeat evaluation on progressively
556 stricter subsets as a sensitivity analysis (full table in

Table 1. Canonical VSR test accuracy (%). Orientation improves little with model size or adaptation compared with several other relation families.

Condition	Overall	Orientation	Depth	Horizontal	Containment	Topology/contact
2B zero-shot	74.0	62.8	68.9	70.2	83.4	80.5
2B structured	68.3	53.3	64.9	66.7	78.7	71.7
2B General LoRA	76.6	62.0	71.1	74.3	87.6	81.0
2B Targeted LoRA	76.5	64.2	70.8	75.1	87.6	81.4
7B zero-shot	80.9	63.5	75.2	84.8	89.3	80.3
7B General LoRA	84.7	65.7	82.3	87.4	92.9	84.4
7B Targeted LoRA	83.9	64.2	81.7	87.1	91.7	85.3
7B Hard-Neg LoRA	84.3	66.4	80.7	87.1	89.9	84.6
7B Projector LoRA	82.9	64.2	78.6	87.1	89.3	85.3
7B Vision+Projector LoRA	83.1	64.2	78.0	87.1	89.9	84.4
MiMo-V2.5 zero-shot	79.5	65.7	74.8	81.9	87.1	84.6

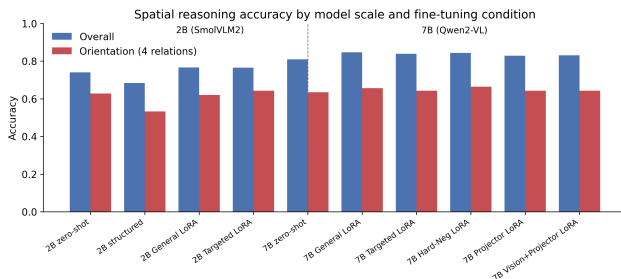


Figure 1. Overall versus orientation performance across the two VLMs and adaptation conditions.

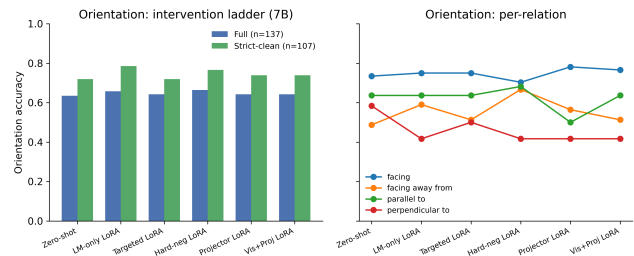


Figure 2. Orientation performance across 7B interventions, including clean-label analysis and per-relation behavior.

the supplementary). The best full-set orientation result is 66.4%; on the automated strictest 107-example subset, the best result rises to 78.5% for 7B General LoRA. Because auditing changes the evaluated sample, these are not matched comparisons against other relation families. Two blind human re-audits under the same taxonomy (binary flag on all 137, eight-class taxonomy on the 48 persistent failures) yield three distinct analyses, reported separately.

Taxonomy-derived strict masks. Each rater’s strict mask raises orientation accuracy without eliminating error—automated 78.5% ($n=107$), human A 77.6% ($n=107$), human B 78.8% ($n=104$)—and 2B structured prompting still shows no gain under any of the three masks (53.3%, 51.4%, 51.0%).

Binary clean/ambiguous masks. The two humans agree substantially on the 137-example binary flag ($\kappa=0.645$, 82.5% agreement) but only moderately with the automated audit ($\kappa=0.36$ on the 48 cases rated by both sources; the automated audit never flagged the remaining 89), and the taxonomy agrees only weakly with the automated labels ($\alpha=0.14$). Under the human binary masks the clean-subset gains largely vanish (best 66.7% and 66.2% vs. the 66.4% full-set best),

tying the gains to the taxonomy-derived masks rather than to the binary flag. Audit reliability is itself model-dependent: two further LLM auditors (GPT-5.6 Sol and Gemini Spark) agree with the humans only weakly ($\kappa=0.34$ and $\kappa=0.03$ on the 137-example flag; Gemini flags 126/137 as clean), so LLM-generated ambiguity labels are sensitivity tools, not authoritative ground truth (App. F).

Consensus analysis. Under the consensus union mask ($n=62$), the trained conditions again improve (7B General LoRA 77.4%) but 7B zero-shot does not (58.1%), and the 2B structured no-gain pattern does not survive this mask (69.3%), consistent with annotator dependence.

The robust conclusion is narrower: taxonomy-derived strict masks raise orientation scores but do not eliminate error across conditions; the audit is exploratory qualitative evidence, not a ruling on label noise (frozen v1 and both human tables in the supplementary).

5. Why Does Orientation Remain Difficult?

5.1. Frozen Representation Probes

We probe frozen Qwen2-VL-7B-Instruct visual representations (no fine-tuning). Mean-pooled ViT and post-merger features support only weak relation decoding for facing versus facing-away, parallel versus perpendicular, and a four-way orientation task. Linear and MLP probes generally remain near majority baselines; preserving patch-level structure yields a suggestive point estimate for parallel/perpendicular (patch-vote 73.5% vs. 64.7% majority), but that test set is small ($n=34$) and the confidence interval overlaps the majority baseline.

To reduce the confound that global features might infer relations without identifying the relevant objects, we additionally extract subject/reference regions and construct grounded visual features from subject, object, difference, and elementwise-product representations. Object-box conditioning with the model’s own grounding predictions and mean-pooled region features does not recover robust orientation decodability. Following standard cautions about diagnostic classifiers [4], we phrase these results as decodability under the tested readouts, not as proof that the information is absent from the network. Linear decodability of visual concepts has been probed in driving-oriented VLMs [13], and directional information in this model family has been inspected via attribution [11]; our probe analysis is not the first of either kind. It differs by targeting VSR’s orientation family with pooled, patch-vote, and object-grounded readouts over the exact conditioning features of the generative model, paired with adaptation and consistency interventions. Detailed probe tables and plots are in the supplementary.

5.2. Visual-Side Adaptation and Object-Centric Decomposition

If the frozen representation were the sole limiting factor, directly adapting visual or projector components might recover orientation performance. For the trained checkpoints evaluated here, it did not: projector LoRA and vision+projector LoRA both produced 64.2% orientation accuracy, below the 65.7% LM-only control, while overall accuracy was nominally lower (82.9% and 83.1% vs. 84.7%; unadjusted $p=0.0043/0.0122$; these effects do not survive a blanket multiple-comparison adjustment and are not load-bearing, App. D). This suggests that the tested low-rank adaptation of the visual pathway does not close the orientation gap.

We also test a box-based two-stage decomposition. Its four-way relation module classifies facing/facing-away/parallel/perpendicular against ground-truth rela-

tion labels at 44–46% cross-validation accuracy (49.9% majority baseline; uniform four-class chance 25%), so the tested relation classifier does not outperform the empirical majority baseline. Consistent with that, the pipeline’s True/False verdicts match the end-to-end control’s correctness on only 55.1–58.3% of the 127 boxed orientation statements (per-relation agreement 25.0–69.4%, weakest for parallel; App. D); in the committed analysis (10 no-box statements scored wrong) the verdicts diverge from the control on 61–65 of the 137 statements, with the control correct and the pipeline wrong in 44–47 of them. These are agreement scores, not ground-truth accuracies: the committed script scored the pipeline against the control’s verdicts, so we report the tested decomposition as an agreement diagnostic that fails to reproduce the generative model’s behavior, not as a direct accuracy comparison (per-example ground-truth scoring is provided by the released script and requires the frozen embeddings, App. G). Two-dimensional box geometry is structurally insufficient for intrinsic facing direction, and the tested region representations do not supply enough signal to close the gap.

5.3. Logical Consistency Reveals a Distinct Failure Mode

We construct complementary statements for strict relation families. A coherent model should return opposite truth values for left/right, front/behind, and facing/facing-away formulations of the same object pair. Complementary-pair and transformation-based consistency evaluation has prior literature, including consistency-oriented training [8]; our contribution is the VSR-specific diagnostic below, built on strict complementary relation pairs with observed verdict-pattern and base-rate analysis. The zero-shot 7B model is consistent on only 58.0% of left/right pairs, 57.0% of front/behind pairs, and 36.9% of facing pairs. The large sparse-MoE zero-shot model does not repair coherence either: it reaches 42.7% facing-pair consistency under the same definition (vs. 36.9% for the 7B zero-shot model; 44.0% on the 100 pairs with parseable verdicts, three excluded for non-conforming outputs), with the same False-defaulting verdict pattern as the smaller models (0 both-True, 56.0% both-False; verdict table in the supplementary). Scale alone therefore does not resolve the coherence failure mode. The observed verdict patterns show what drives this: a False-defaulting bias that yields both-False verdicts on 63.1% of pairs and no both-True verdicts (0/103), with expected consistency under verdict independence at 30.2%, below the observed 36.9% (verdict table in the supplementary): the dominant phenomenon is a refusal

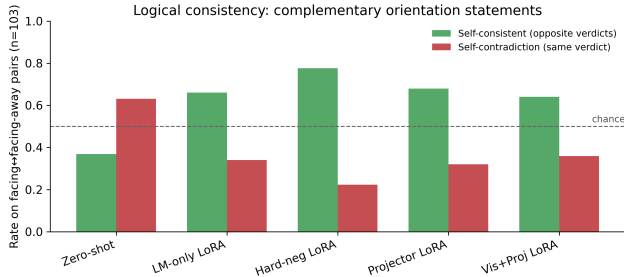


Figure 3. Self-consistency on complementary facing/facing-away relations across 7B conditions.

to affirm either proposition, rather than a preference for contradictory affirmations.

LM-only LoRA raises facing consistency to 66.0% and front/behind to 70.7% (pooled strict-family significantly better than zero-shot, $p < 0.0001$); hard-negative training raises facing consistency to 77.7% with facing accuracy identical at 68.9% for both, although the additional pooled consistency effect over LM-only is not significant ($p = 0.29$). The key result is separability: identical facing accuracy coexists with substantially different facing consistency; we do not claim a general coherence algorithm. Training also changes the failure mode rather than eliminating it: both-False facing verdicts drop from 63.1% to 9.7% (LM-only) while both-True contradictions rise from 0% to 24.3%, a disclosed specialization trade-off.

6. Cross-Dataset Transfer to SITE

We evaluate the frozen 7B base model and the VSR-trained 7B General LoRA adapter on the same 2,591 SITE image examples using SITE’s native multiple-choice protocol [15], with no SITE training. Because the pre-specified decision rule resolved to branch 2, this adapter run is a post-protocol transfer check rather than a pre-specified step (protocol and outcome in the supplementary). The model is strong on the official spatial-relationship-reasoning category: 75.1% raw accuracy and 59.2% CAA. Applying the VSR-trained adapter does not transfer cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$; 52 examples fixed vs. 87 broken). VSR gains therefore transfer poorly and can induce negative transfer; we do not equate benchmark-specific fine-tuning gains with general spatial-reasoning improvement.

The pre-specified, non-official orientation-vocabulary slice is much weaker in the aggregate (48.3% raw / 24.0% CAA vs. 75.1% / 59.2% for the official category) and shows no positive transfer under

Table 2. SITE cross-dataset transfer (images). Entries are raw accuracy / CAA (%).

Subset	Zero-shot	VSR-LoRA	Δ
All images	54.2 / 31.1	53.8 / 30.5	-0.4
Official spatial-rel.	75.1 / 59.2	71.6 / 53.4	-3.5
Orient. heuristic	48.3 / 24.0	48.7 / 24.5	+0.4

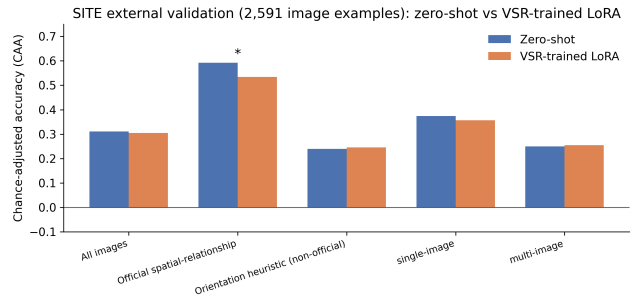


Figure 4. SITE CAA for zero-shot and VSR-trained LoRA. The official spatial-relationship category degrades significantly under transfer.

the adapter (48.3%→48.7%, $p = 0.70$). Because the slice is keyword-derived and heterogeneous (43% of its examples come from multi-view/cross-image reasoning, only 24% from the spatial-relationship category), we treat it as exploratory evidence rather than replication of the VSR construct. A composition-confound analysis of the 2,591 zero-shot predictions (supplementary §E) supports this caution: after controlling for official category, source dataset, modality, and number of options, the orientation flag is not significantly associated with correctness (odds ratio 0.84, 95% CI [0.60, 1.16], $p = 0.29$), although it remains descriptively 7.5pp lower within the official spatial-relationship category itself (71.3% vs. 78.8%, $n=488/505$). A post-hoc high-precision subset restricted to VSR-construct terms (facing, facing away, parallel, perpendicular; $n=177$) yields 44.1% raw (95% CI [37.0, 51.4]; CAA -2.3%) and is reported as exploratory only. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck; the transfer results stand on the official category, where the model is strong and the adapter demonstrably does harm.

7. Discussion

Orientation is not spatial reasoning broadly. The strongest conclusion is deliberately narrow. The 7B model is strong on several VSR families and SITE’s official spatial-relationship category. The persistent weakness concerns the object-relative orientation family in VSR; on SITE, the orientation-vocabulary slice

595 is analyzed as exploratory evidence rather than as inde-
596 pendent confirmation. This framing is consistent with
597 the original VSR observation [7] while extending it to
598 modern generative VLMs and a broader intervention
599 analysis.

600 Decodability and coherence both matter, as diagnos-
601 tics. The probes show that orientation is not robustly
602 decodable under the tested frozen-feature readouts,
603 while consistency experiments expose large logical fail-
604 ures even for relations answered above chance. Neither
605 result alone identifies a complete causal mechanism,
606 and both are diagnostic rather than causal-localization
607 claims. Together they support a mixed account in
608 which orientation information is difficult to access reli-
609 ably under the tested readouts and relational decisions
610 are not always propagated coherently across comple-
611 mentary formulations.

612 Task adaptation can specialize rather than generalize.
613 General LoRA improves VSR overall accuracy substan-
614 tially, but its SITE transfer is neutral overall and signif-
615 icantly negative on the official subset. Hard negatives
616 improve a targeted coherence behavior with relation-
617 specific tradeoffs. Benchmark-specific fine-tuning gains
618 should therefore not be equated with general spatial-
619 reasoning improvement.

620 8. Limitations

621 Statistical power is limited at the scale of the orien-
622 tation diagnostic. The VSR orientation family con-
623 tains only 137 test examples, and the smallest per-
624 relation subset, perpendicular, has $n=12$; perpendicu-
625 lar results are therefore reported descriptively through-
626 out. The Wilson 95% confidence intervals on the
627 family-level point estimates are correspondingly wide—
628 62.8% [54, 70], 63.5% [55, 71], 65.7% [57, 73], and 66.4%
629 [58, 74] for 2B zero-shot, 7B zero-shot, 7B General
630 LoRA, and 7B Hard-Neg LoRA ($n=137$ each)—so
631 the trained conditions’ 62–66% band should be read
632 as a set of statistically overlapping point estimates
633 rather than a fine-grained ranking. Several smaller
634 point estimates have confidence intervals that include
635 the majority baseline or chance: the patch-vote paral-
636 lel/perpendicular probe (73.5%, $n=34$; 95% CI [57, 85]
637 vs. the 64.7% majority), the perpendicular subset for
638 the trained 7B conditions (41.7%, $n=12$; [19, 68] vs.
639 50%), and the 7B zero-shot facing-away subset (48.7%,
640 $n=39$; [34, 64]). Paired McNemar tests on the ori-
641 entation family rest on small discordant-pair counts
642 and have limited power: the non-significant projector
643 and vision+projector orientation comparisons ($p=0.86$,

644 $p=0.85$) indicate an absence of detectable effect, not ev-
645 idence of equivalence. Where the paper draws conclu-
646 sions, it relies on the full-set paired tests and the Holm-
647 adjusted battery in the supplementary rather than on
648 these small- n point estimates alone.

649 The clean-label audit’s exclusion sets originate from
650 an automated LLM audit: two independent human re-
651 audits reached only moderate per-item agreement with
652 it ($\kappa=0.36$ on the clean/ambiguous flag; Krippendorff’s
653 $\alpha=0.14$ on the eight-class taxonomy) while agreeing
654 strongly with each other on the failure taxonomy
655 ($\alpha=0.81$); both reproduced the qualitative conclusions
656 under their own taxonomy-derived strict masks, so the
657 sets are annotator-dependent, and the frozen table is re-
658 ported as v1 with both human re-audit tables alongside.
659 The SITE orientation slice is keyword-derived rather
660 than an official benchmark category; a composition-
661 confound analysis shows its aggregate score is largely
662 explained by task composition, so we do not treat SITE
663 as independent confirmation of the VSR orientation
664 bottleneck. Our probing methods test accessible decod-
665 ability under specific linear/MLP and pooling choices;
666 failure to decode does not prove information is ab-
667 sent. Visual-side experiments use LoRA rather than
668 full end-to-end retraining. We study two differently
669 sized VLMs from different families plus a large sparse-
670 MoE third model; this is a cross-model comparison,
671 not a controlled scale experiment. All trained condi-
672 tions are single-checkpoint runs: the paired McNemar
673 tests quantify disagreement on the fixed test examples,
674 not training-run variability, and these single-seed re-
675 sults do not estimate it (a multi-seed extension is pre-
676 pared in scripts/run_seed_variance.py). The MiMo-
677 V2.5 evaluation is served through a public API: 3.1%
678 of its outputs did not conform to the one-word for-
679 mat and are counted as wrong (invalid outputs are
680 never guessed), and 294 responses contained chain-of-
681 thought text alongside the verdict; both are disclosed
682 in the committed predictions (App. A). The MiMo-
683 V2.5 row was evaluated on 2026-08-12 through the
684 OpenCode Go gateway; API-served models can be up-
685 dated without notice, so this row’s reproducibility de-
686 pends on the serving snapshot (evaluation date, model
687 id, and per-request token accounting are recorded in
688 results/mimo/). The SITE evaluation is image-
689 only; SITE video reasoning remains future work. Fi-
690 nally, several interventions are evaluated as negative
691 results, so lack of improvement should be interpreted
692 for the tested configurations rather than as impossibil-
693 ity claims about all prompting, adaptation, or architec-
694 tural methods.

9. Conclusion

Aggregate spatial accuracy hides a persistent orientation bottleneck in the studied VLMs: a larger VLM and standard adaptation substantially improve several spatial families while leaving object-relative orientation resistant, and a large 2026 sparse-MoE model reproduces the pattern. Auditing, probing, visual-side adaptation, and decomposition constrain simple explanations; complementary-relation tests show accuracy and coherence are distinct axes that both remain weak in the large model. SITE transfer shows VSR-specific adaptation can negatively transfer. Evaluation should report relation-specific performance, transfer, and logical coherence rather than aggregate accuracy.

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